

Azeem Shaik

SOFTWARE ENGINEER

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SUMMARY

Software Engineer with 3+ years of experience at EPAM Systems and MongoDB, developing backend services and React applications. Work spans API development, database performance, asynchronous processing, and automated delivery. Applied AI experience includes integrating LLM features into engineering workflows and evaluating document retrieval, cited answers, and access controls.

EDUCATION

MS, Computer and Information Science | University of Alabama at Birmingham | May 2026

PROFESSIONAL EXPERIENCE

MongoDB | Software Engineer | US | June 2025 - Present

- Optimized 10+ MongoDB queries and Node.js/Express API workflows, reducing redundant database calls by 30% and improving API response times by 20%.
- Cut management-console query time from 180 ms to 110 ms after reviewing MongoDB execution plans; revised compound indexes and aggregation stage order to reduce document scans.
- Added idempotency checks and bounded retries to Kafka consumers processing background updates; replayed 10,000 synthetic events with injected failures to verify concurrent processing and recovery after lost acknowledgments.
- Delivered three management-console workflows using React, TypeScript, and TanStack Query, working with design to define loading states and failure behavior before release.
- Standardized deployment of three services on AWS EKS using Kubernetes and Terraform; added release checks covering 22 JUnit backend regression cases and verified rollback behavior in staging.
- Integrated an OpenAI-based incident-summary feature into an engineering console, reducing median summary preparation time from 12 to 5 minutes across 40 paired trials; preserved source references and required engineer approval before saving.

EPAM | Software Engineer | IN | September 2022 - July 2024

- Developed 12 Spring Boot REST endpoints for submission, approval, and status tracking, using PostgreSQL and documenting OpenAPI contracts; integrated OIDC sign-in and OAuth 2.0 access-token validation, enforcing RBAC.
- Replaced repeated reference-data queries with Redis caching and invalidation on record updates, reducing database calls from five to two per request.
- Connected React screens to backend APIs and coordinated form edits and approval state through Redux Toolkit, resolving eight integration defects before the client demonstration.
- Shortened Jenkins regression execution from 28 minutes to 17 minutes by reorganizing JUnit and Mockito checks; worked with QA to reproduce intermittent failures and distinguish application defects from pipeline issues.

PROJECTS

Versioned Workflow Execution Platform | TypeScript, Node.js, PostgreSQL, Next.js

- Built REST APIs and PostgreSQL data models for conditional workflows, versioned definitions, and execution histories; developed a Next.js editor that kept active runs tied to their original configuration.
- Covered 24 editing and execution scenarios with Jest and Playwright; deployed the Docker application to GCP Cloud Run through GitHub Actions, with infrastructure defined in Terraform.

Access-Controlled Document Retrieval Application | Python, MongoDB Atlas Vector Search, OpenAI API

- Built document ingestion and embedding pipelines on MongoDB Atlas, using vector retrieval to supply context for generated answers; enforced document permissions within the search function using authenticated identity.
- Exposed document search through OpenAI tool calling and returned cited answers through validated structured outputs; added explicit handling for questions without sufficient supporting evidence.
- Compared keyword and vector retrieval across 100 labeled questions, measuring recall@5 of 72% and 84%, respectively; used pytest to check citation validity, malformed tool arguments, and unauthorized document access.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, Bash

Backend: Spring Boot, Node.js, Express.js, REST APIs, OpenAPI, Microservices, Event-Driven Architecture, Kafka, OAuth 2.0, OpenID Connect (OIDC), Role-Based Access Control (RBAC)

Frontend: React, Next.js, Redux Toolkit, TanStack Query, Tailwind CSS

Databases & Search: MongoDB, MongoDB Atlas, Atlas Vector Search, PostgreSQL, Redis, Mongoose, Data Modeling, Database Indexing, Query Optimization

Cloud & DevOps: AWS (S3, EC2, Lambda, EKS), GCP (Cloud Run, Cloud Storage, Pub/Sub), Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Prometheus, Grafana

Testing: JUnit, Mockito, Jest, pytest, Playwright, Integration Testing, End-to-End Testing, Performance Testing

Applied AI: OpenAI API, Embeddings, Retrieval-Augmented Generation (RAG), Tool Calling, Structured Outputs, LLM Evaluation